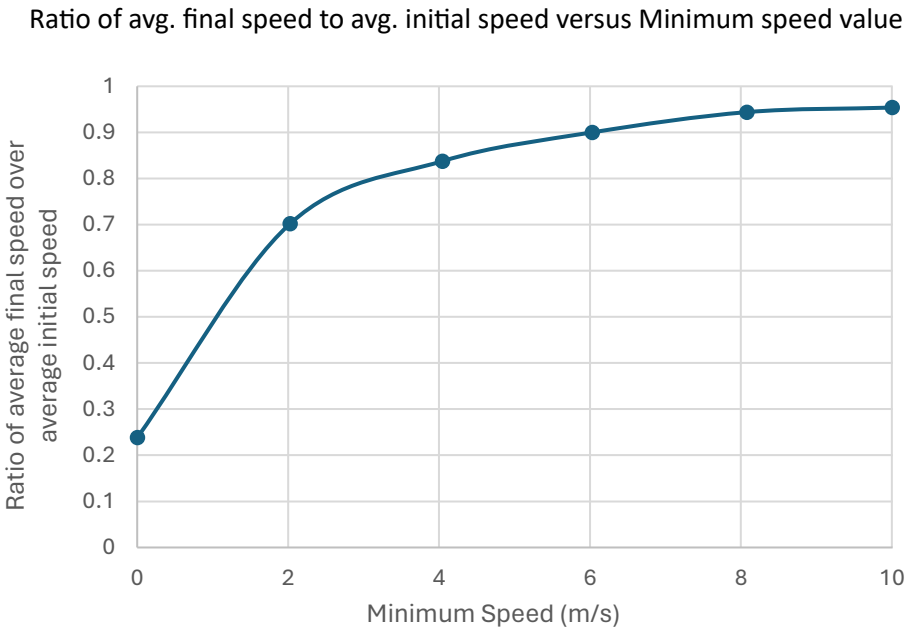
Félix Arnold

ECE 6110 Lab Assignment 3
Mobility Models

Part 2: Evaluation of Minimum Speed Effect in Random Waypoint Mobility

In this part, we study the long-term average speeds with the random waypoint model.

We can observe the ratio of average final speed to average initial speed compared to the minimum speed value.



Thanks to this plot for the random waypoint model, we can observe that when the minimum speed increases from 0 to 10 $m.s^{-1}$, the ratio of average final speed to average initial speed also increases. However, if this increase has a large impact with a very small minimum speed, after this initial phase the gain is much lower with bigger minimum speed (and as such with higher average initial speed). Thus, there is an optimal range for the minimum initial speed to have a real significant impact on the long-term average speed, after which the gain is only marginal.